



FABRIC MANUFACTURING II

Part: B

Course Code: FE 351-0723

Introduction to Warp Knitting

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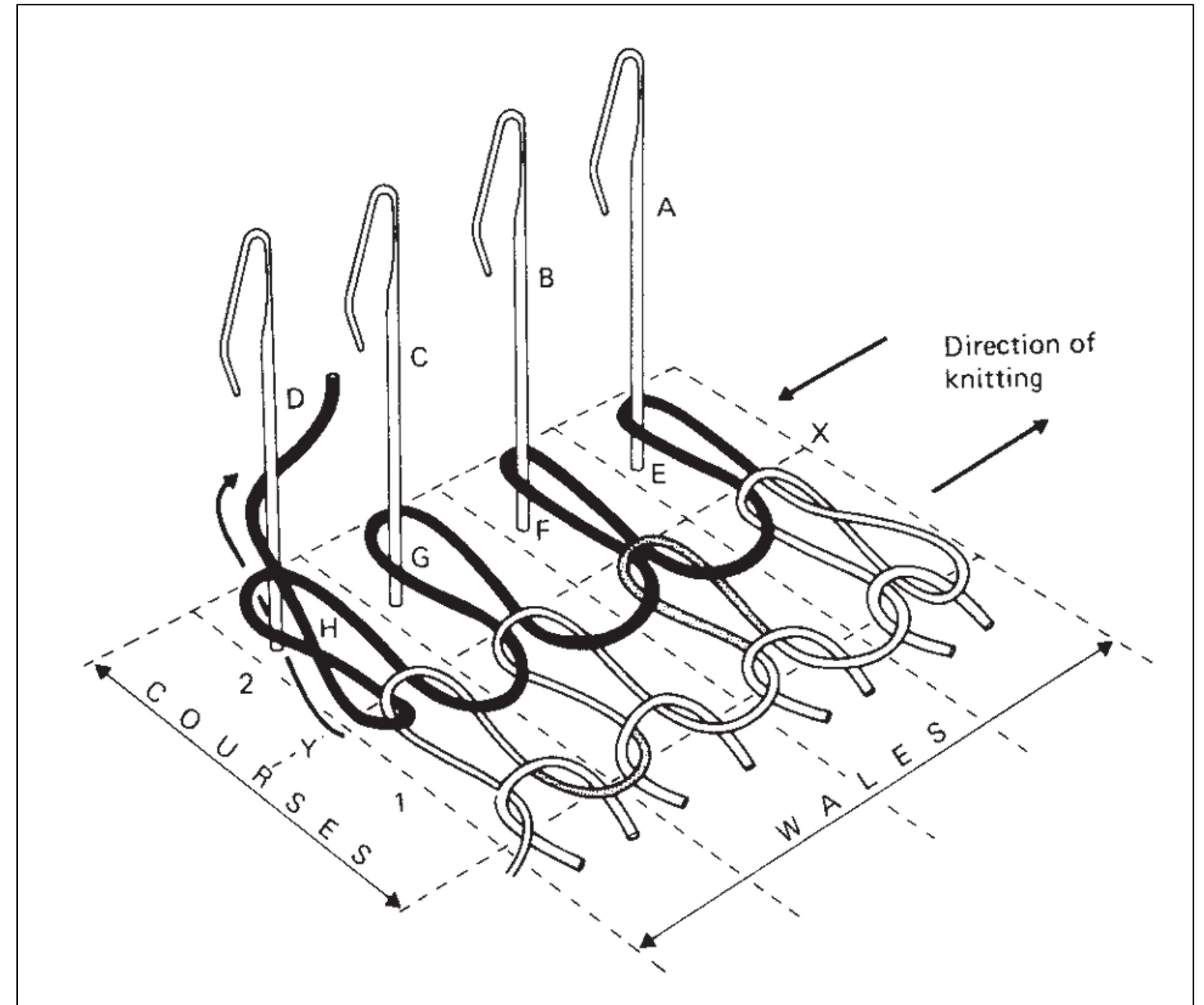
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TYPES OF KNITTING

Weft Knitting:

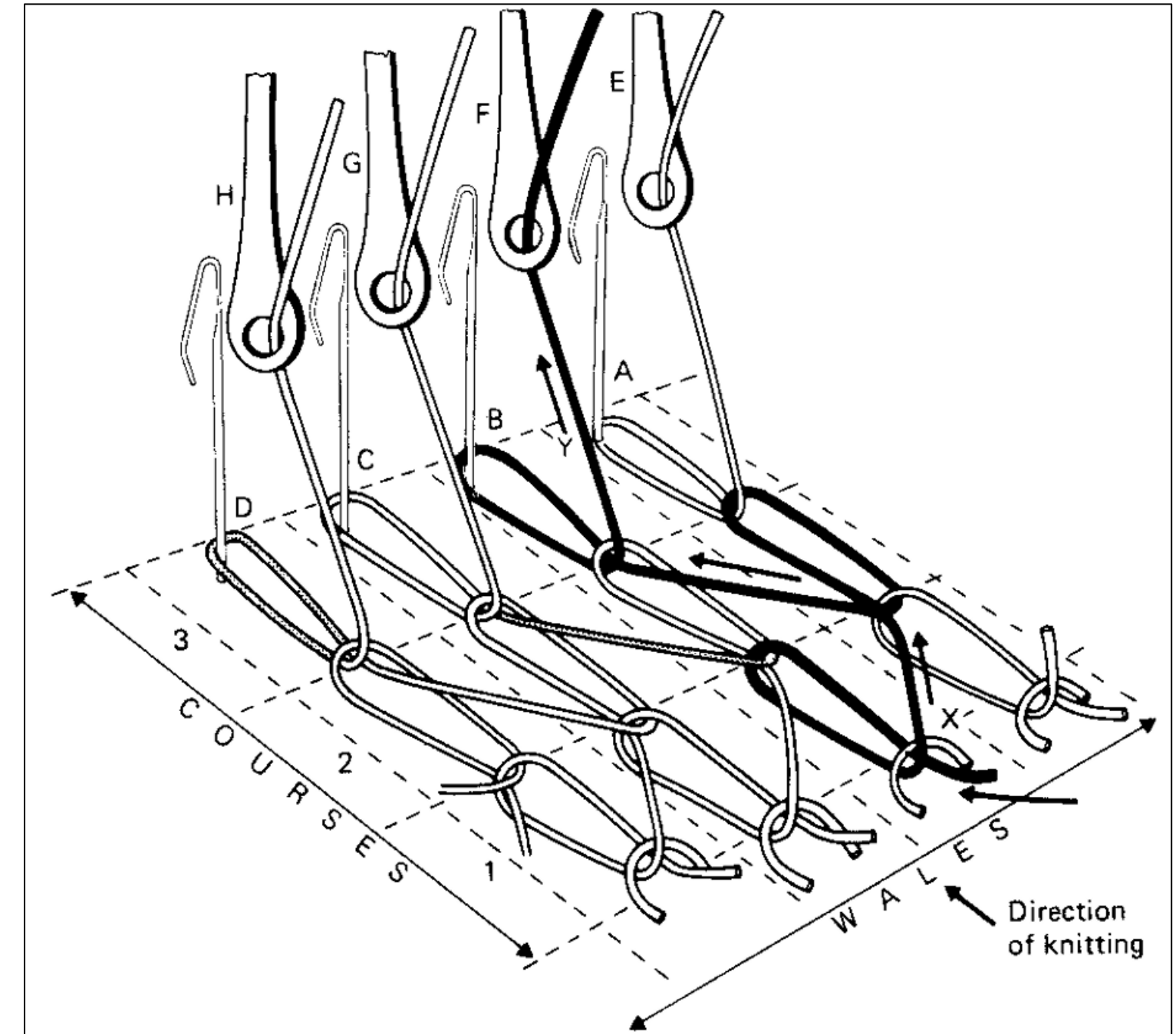
A weft knitted structure is a *horizontal row of loops* can be made using one thread and thread runs in horizontal direction.



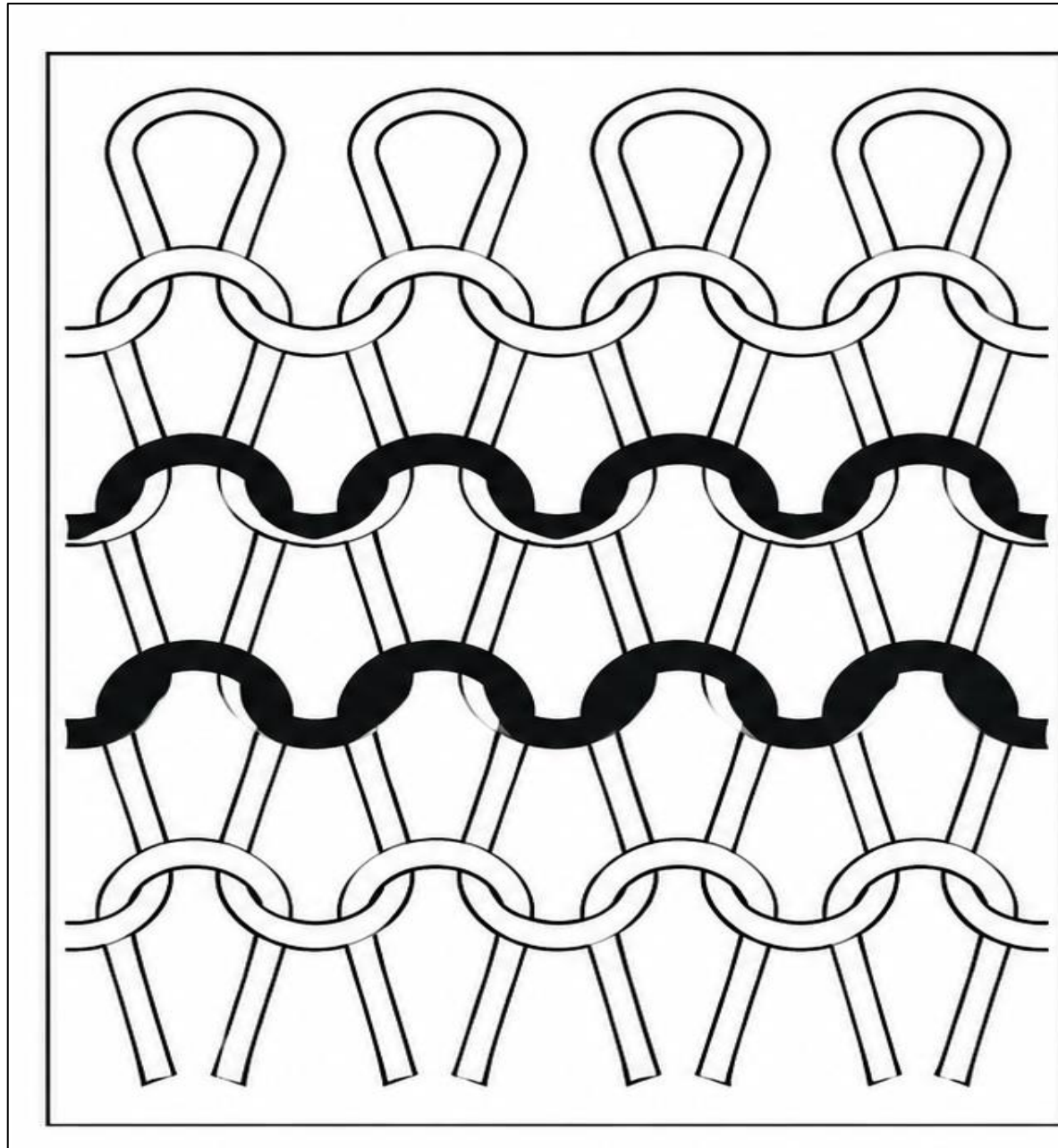
TYPES OF KNITTING

Warp Knitting:

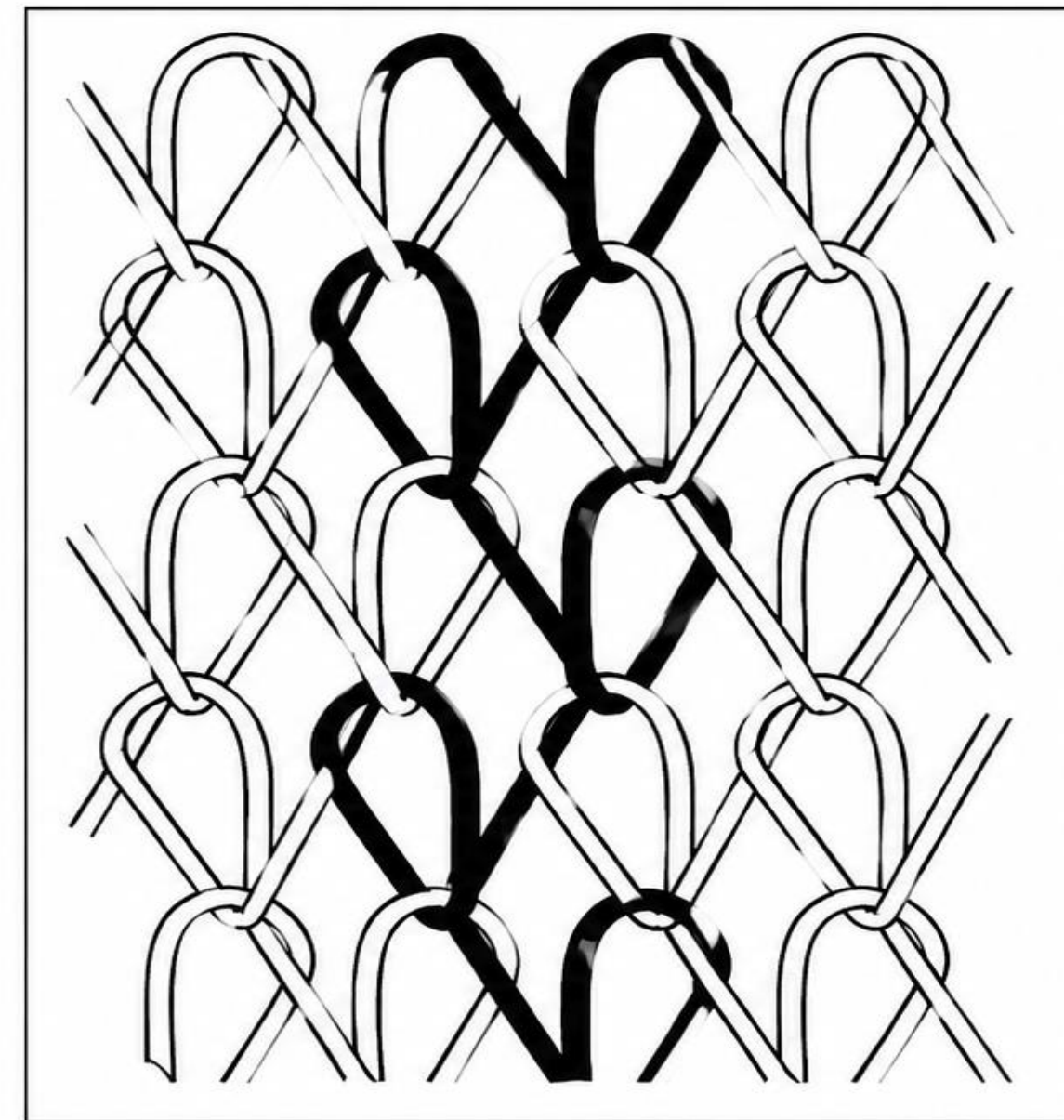
In warp knitted structure *each loop in horizontal direction is made from a different thread* and the *number of threads* used to produce such fabric is *equal to the number of loops in a horizontal row*.



TYPES OF KNITTING (CONTINUED)



WEFT KNITTED
STRUCTURE

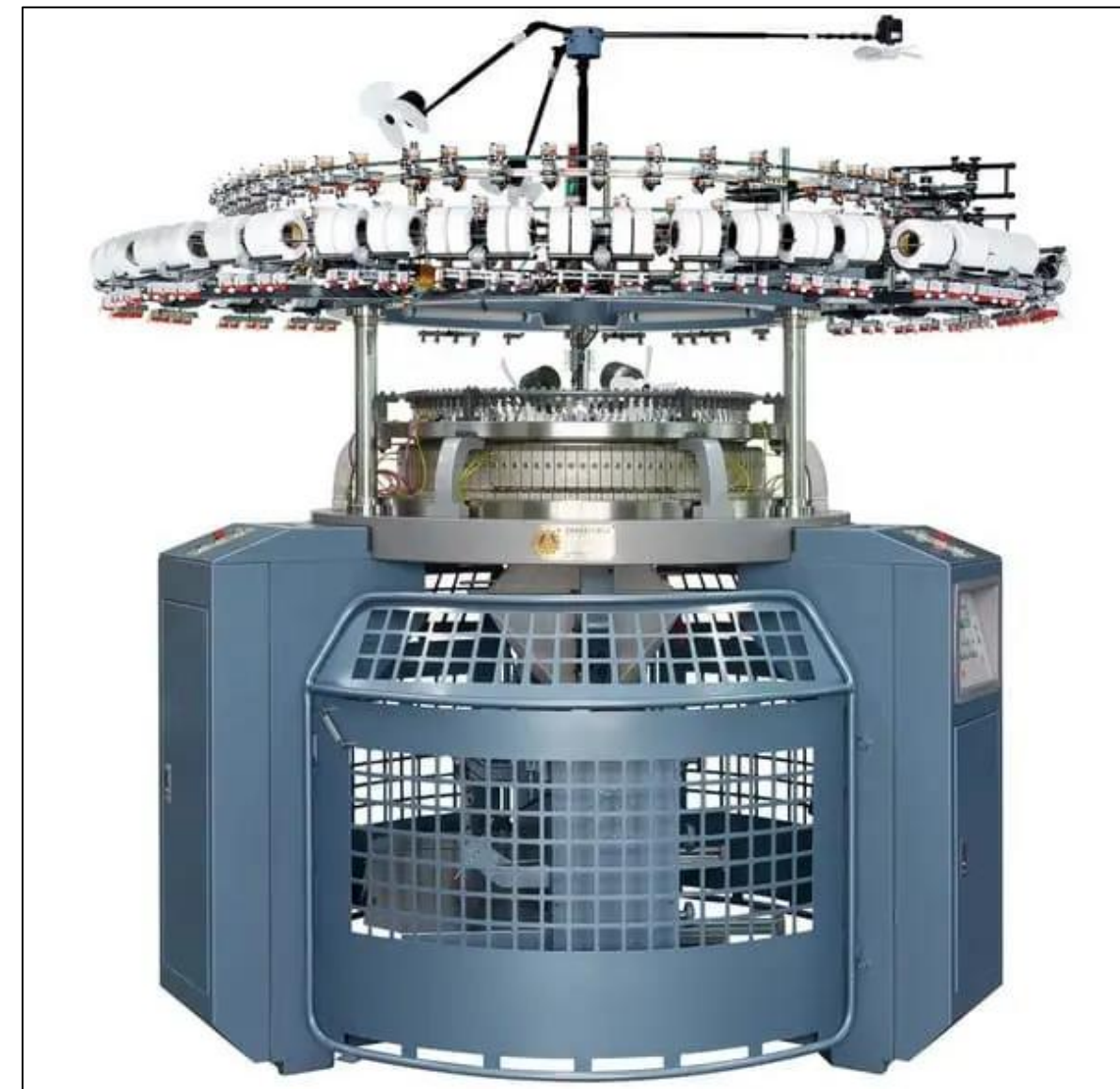


WARP KNITTED
STRUCTURE

TYPES OF KNITTING (CONTINUED)



TRICOT WARP KNITTING MACHINE



SINGLE JERSEY CIRCULAR KNITTING MACHINE

TYPES OF KNITTING (CONTINUED)

Warp Knitting	Weft Knitting
1. Loop formation in the fabric takes place wale wise.	1. Loop formation takes place course wise.
2. Yarns are supplied from a beam in the machine.	2. Yarns are supplied from cones in the machine.
3. Yarns produced from synthetic fibre are mostly used.	3. Yarns produced from natural fibre are mostly used.
4. Antistatic lubricants is used to prevent static electricity buildup (a big issue since many yarns run in parallel at high speeds).	4. Special low-viscosity knitting oil is used which reduces friction between yarn, needles, and machine parts.
5. Design change process (link in the patterned drum or pattern disc) of the machine is less time consuming.	5. Design change (needle and cam) process of the machine is more time consuming.

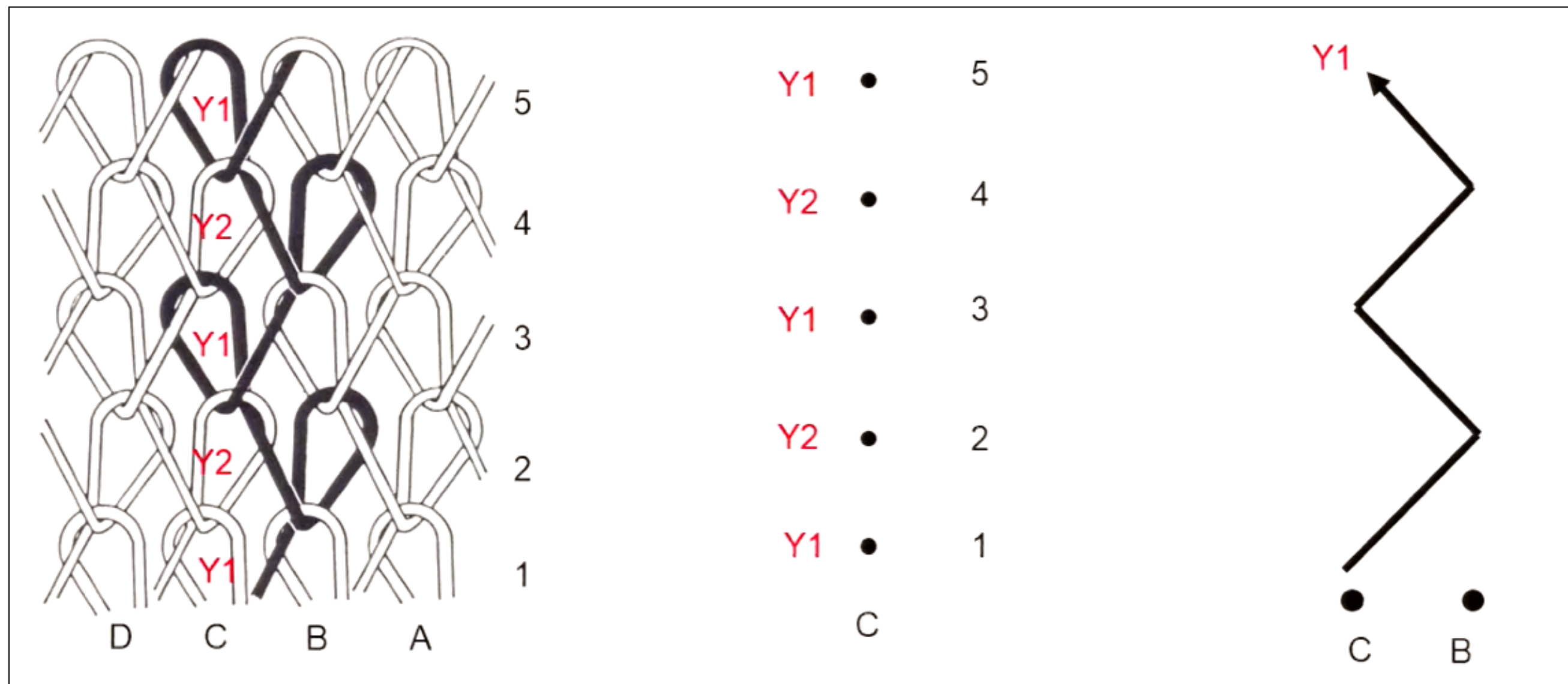
TYPES OF KNITTING (CONTINUED)

Warp Knitting	Weft Knitting
6. Knitting elements of the machine act unitely.	6. Knitting elements act independently.
7. Fabrics have unlimited application, such as inner and outer wears, industrial, technical, and domestic.	7. Limited application, mostly apparel.
8. Latch, bearded or compound needles are used in warp knitting machine.	8. Usually latch needles are used on weft knitting machines.
9. Both Fabrics and machines are expensive.	9. These are cheaper than warp knitting.
10. All the loops of a course are formed exactly at the same time.	10. The loops of a course are formed sequentially.

TYPES OF KNITTING (CONTINUED)

Warp Knitting	Weft Knitting
11. More stable fabric dimensionally.	11. Less stable.
12. The fabric is more suitable for cutting and sewing.	12. Less suitable.
13. Every needle is required to have individual yarn feed.	13. A single yarn is sufficient for all the needles.

WARP KNIT STRUCTURE



A-D Needles, 1-5: Courses, Y: Yarn End

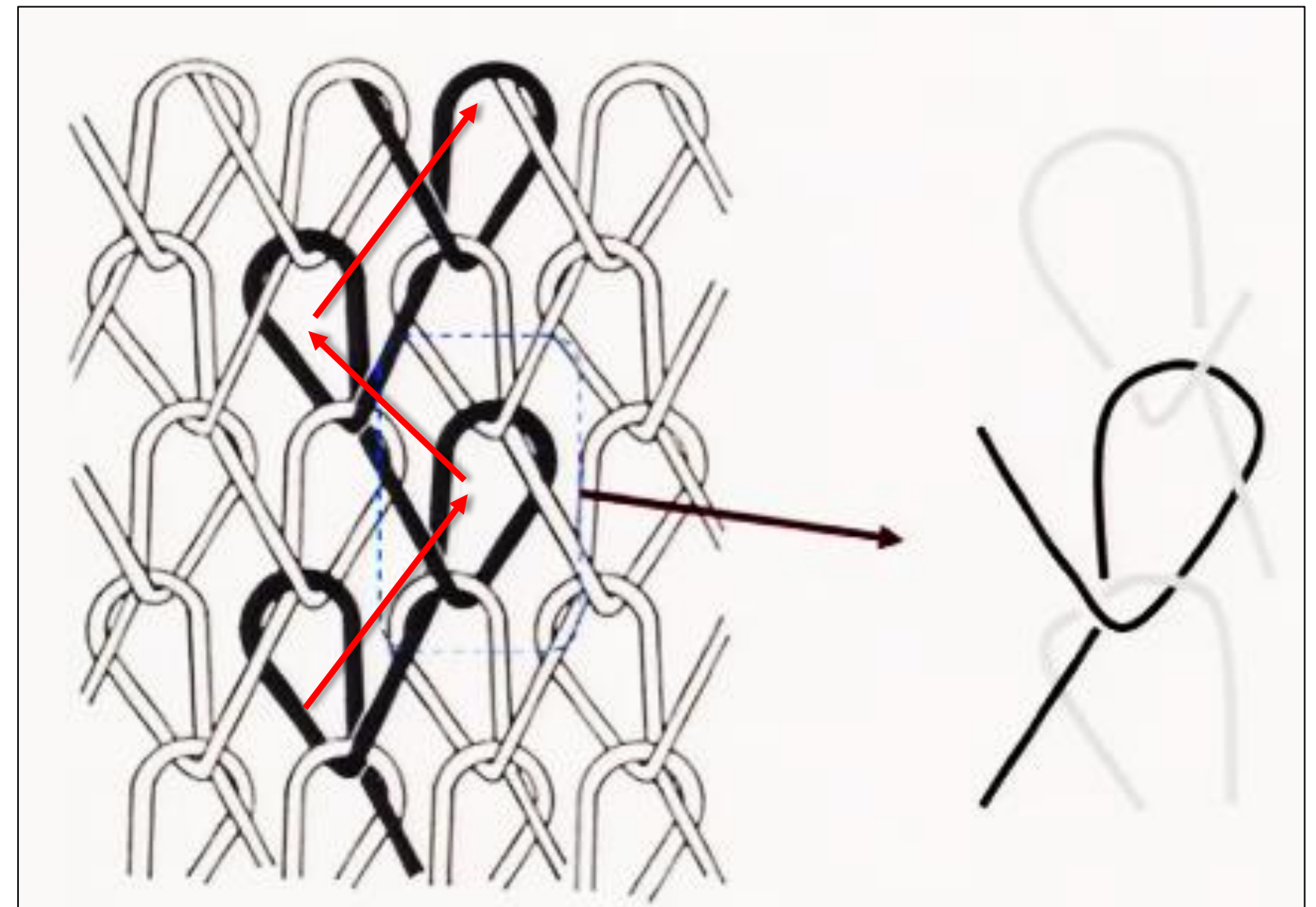
WARP KNIT STRUCTURE

Weft Knitting:

- A loop joins with the loop beside it (left and right side).
- Loops are connected in the same row.

Warp Knitting:

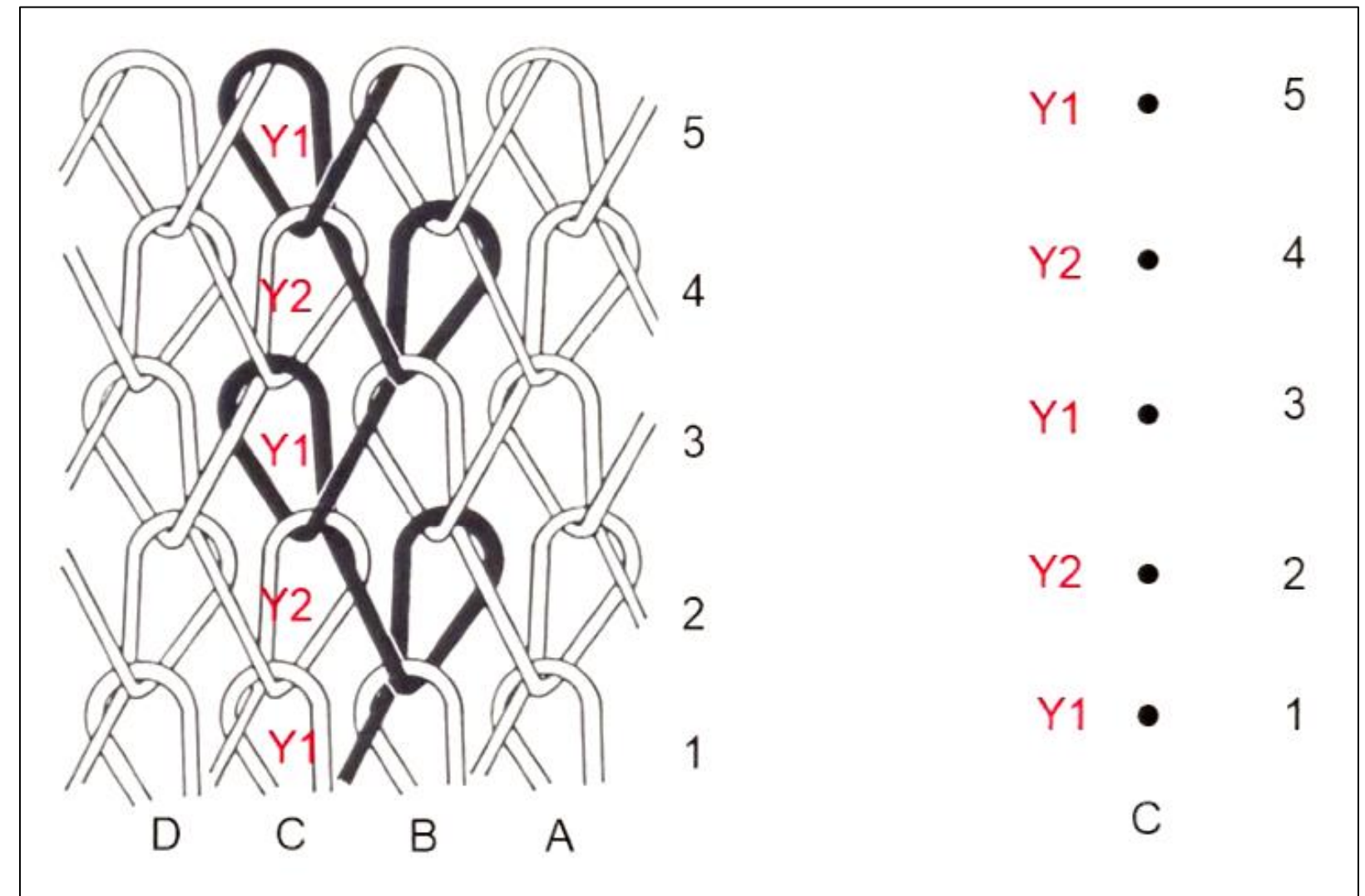
- A loop joins with the loop above and below it.
- Loops are connected between different rows.
- A loop does not join with the left or right loop.



WARP KNIT STRUCTURE

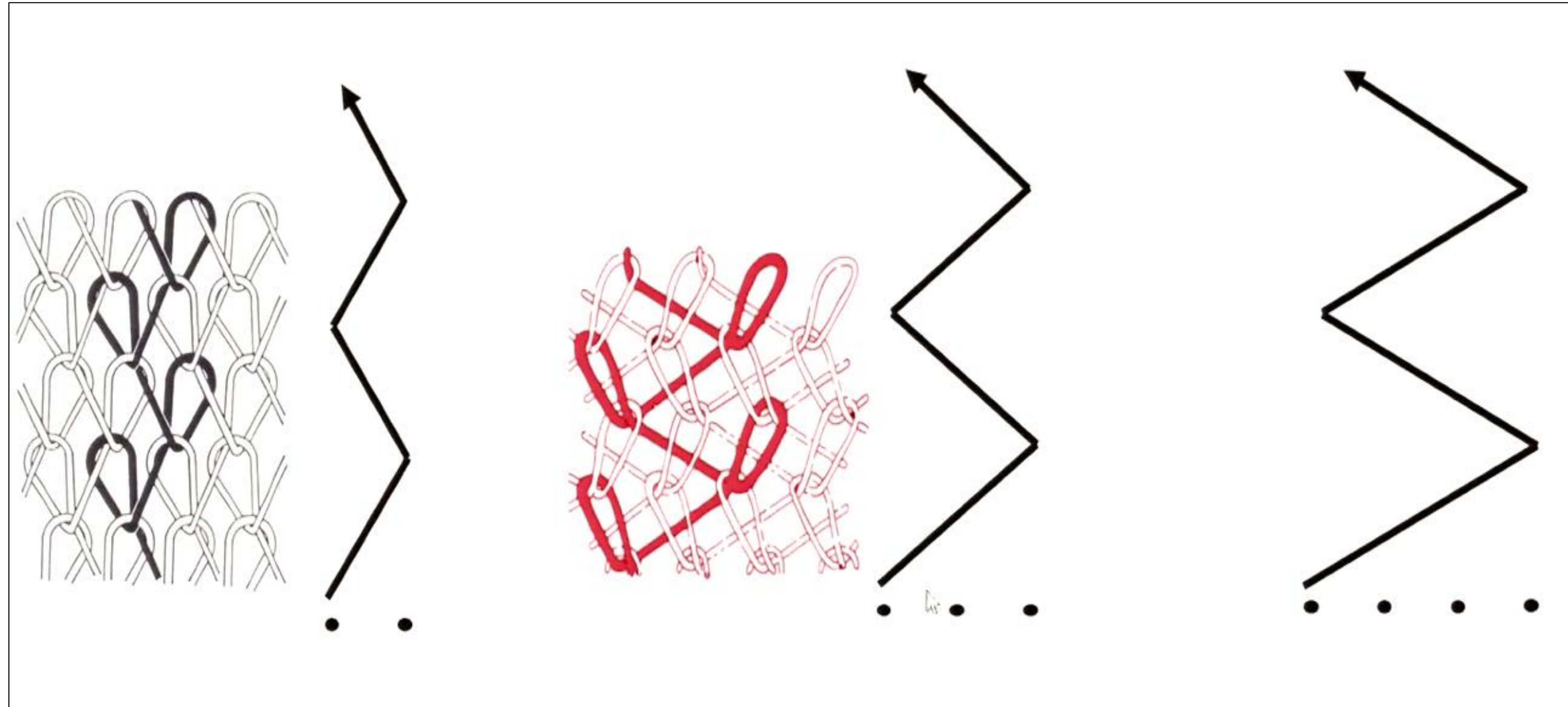
Each wale is created by one needle. Same needle (C) sometimes getting Yarn Y1 and another time Yarn Y2 for loop formation.

Yarn (Y) is moving towards the length direction hence termed as Warp Knitted Structure.

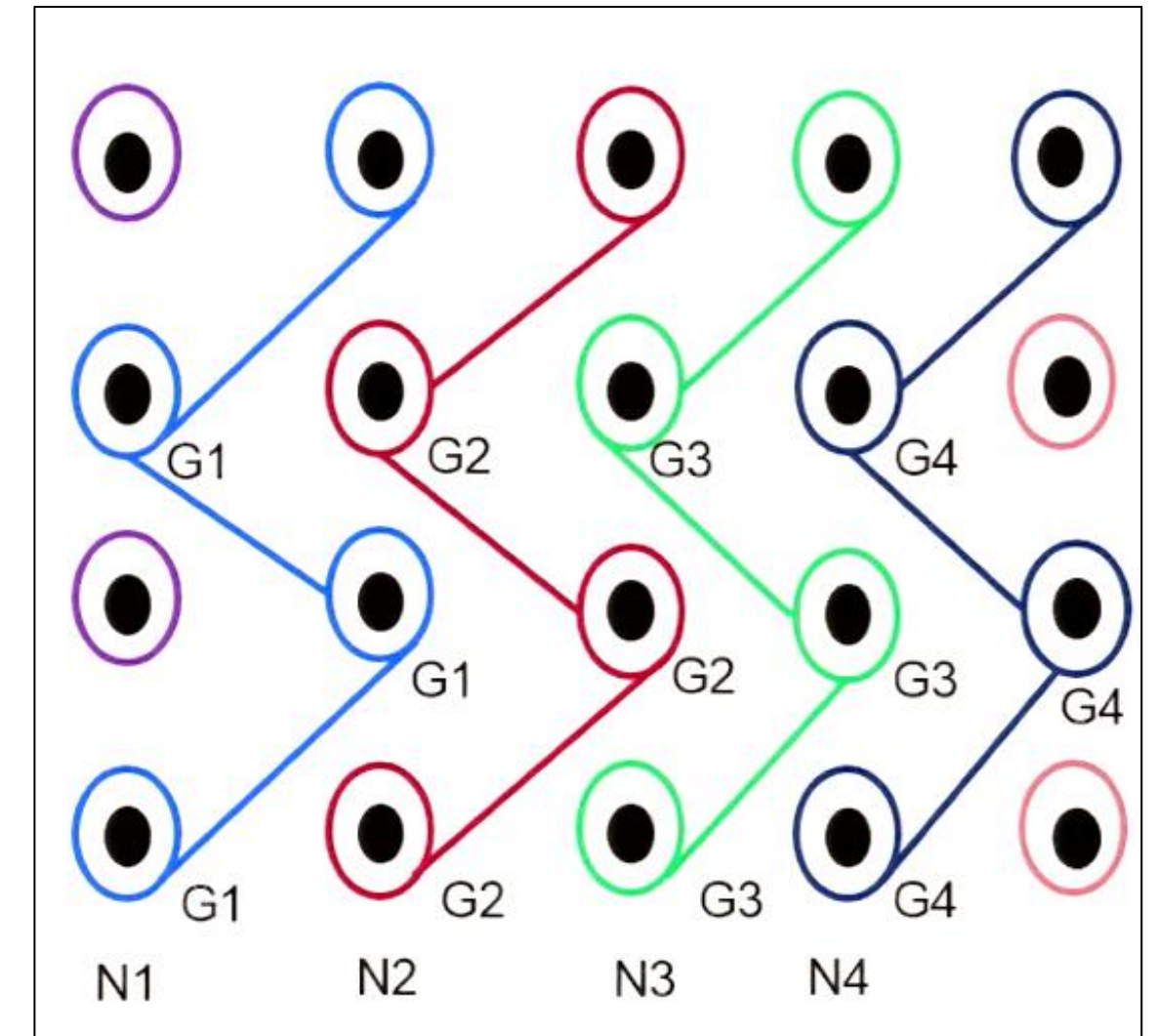
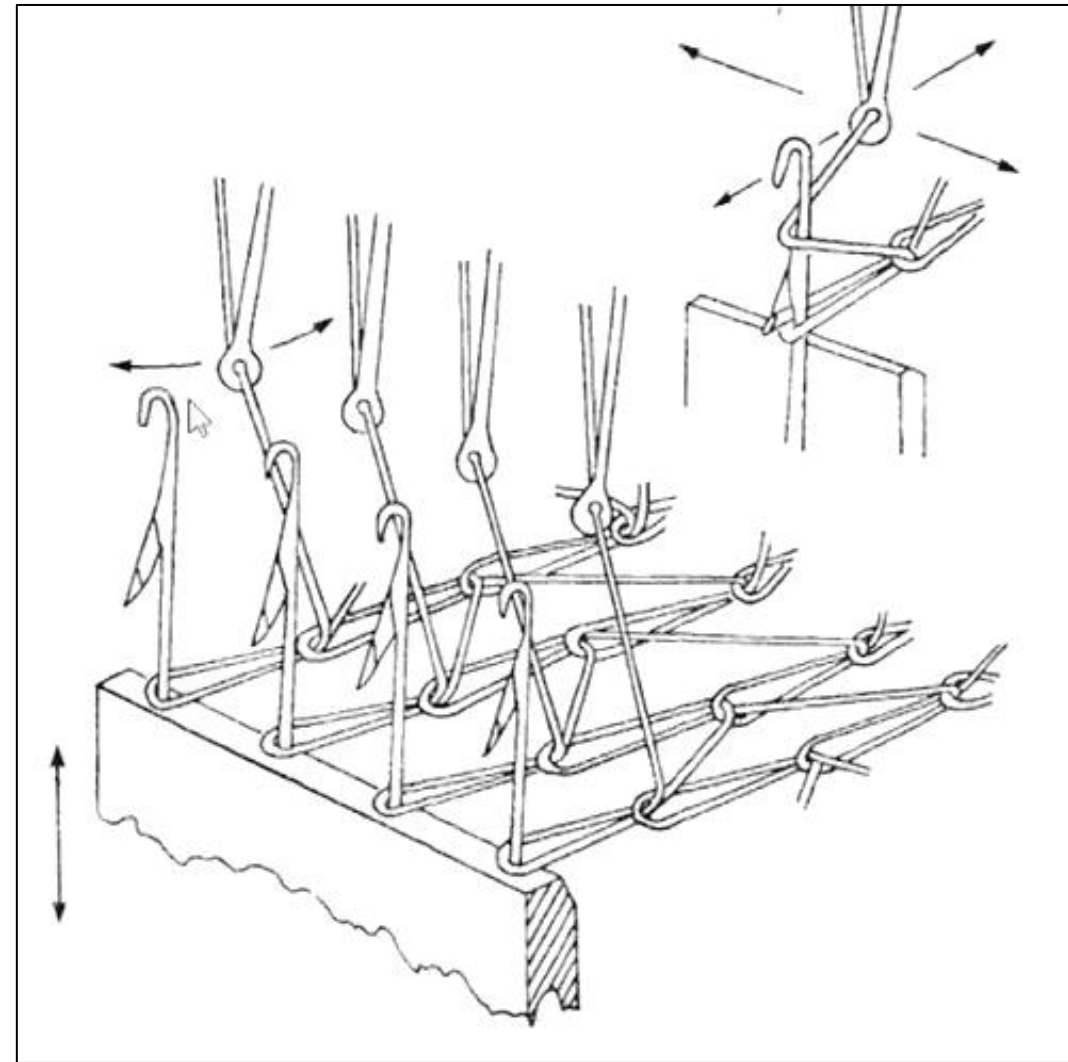
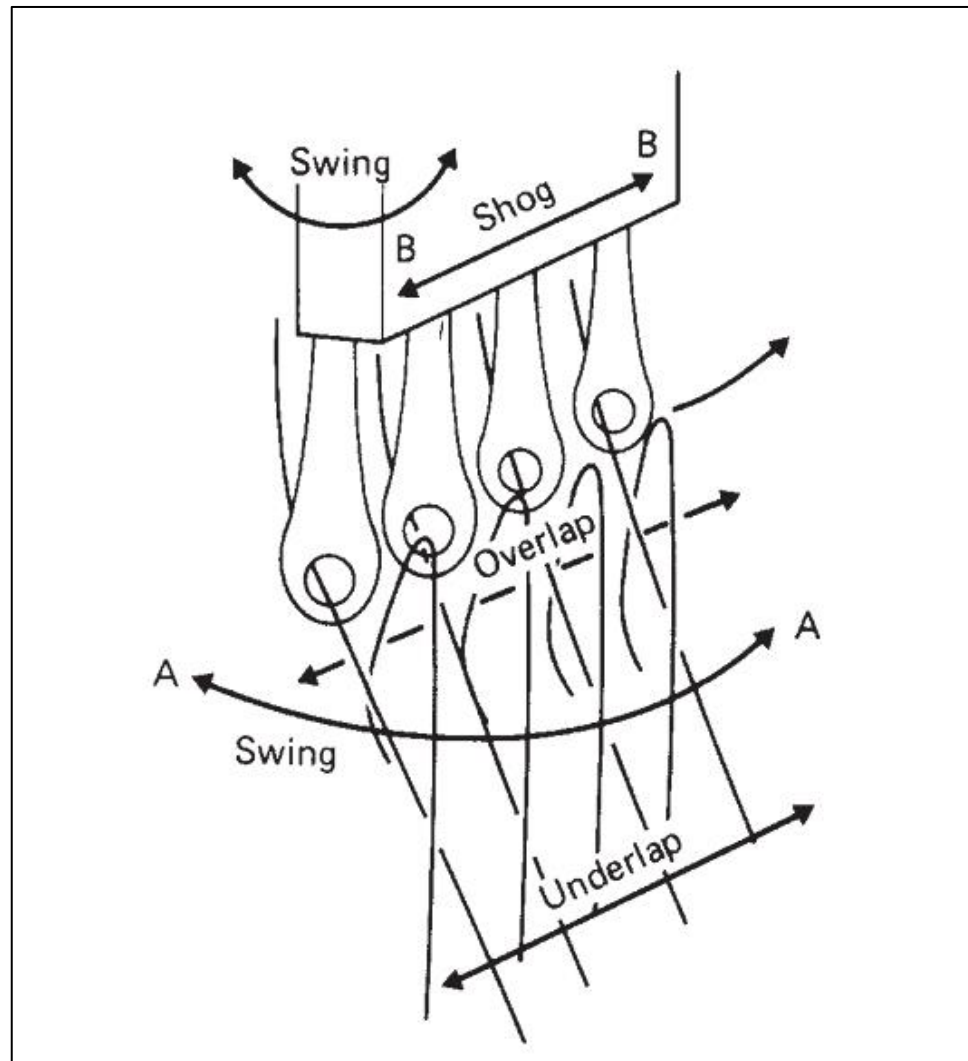


A-D Needles, 1-5: Courses, Y: Yarn End

WARP KNIT STRUCTURE

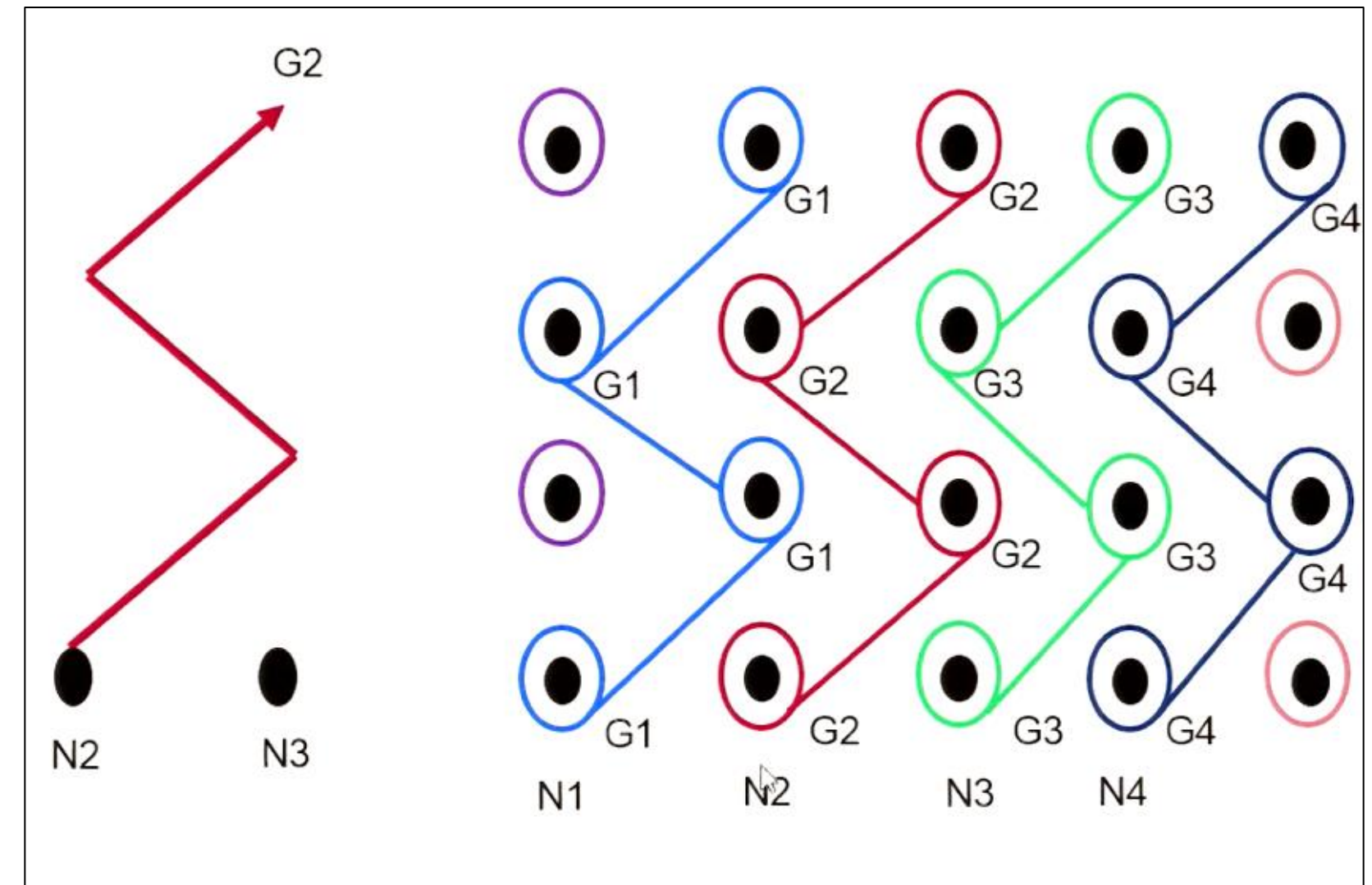
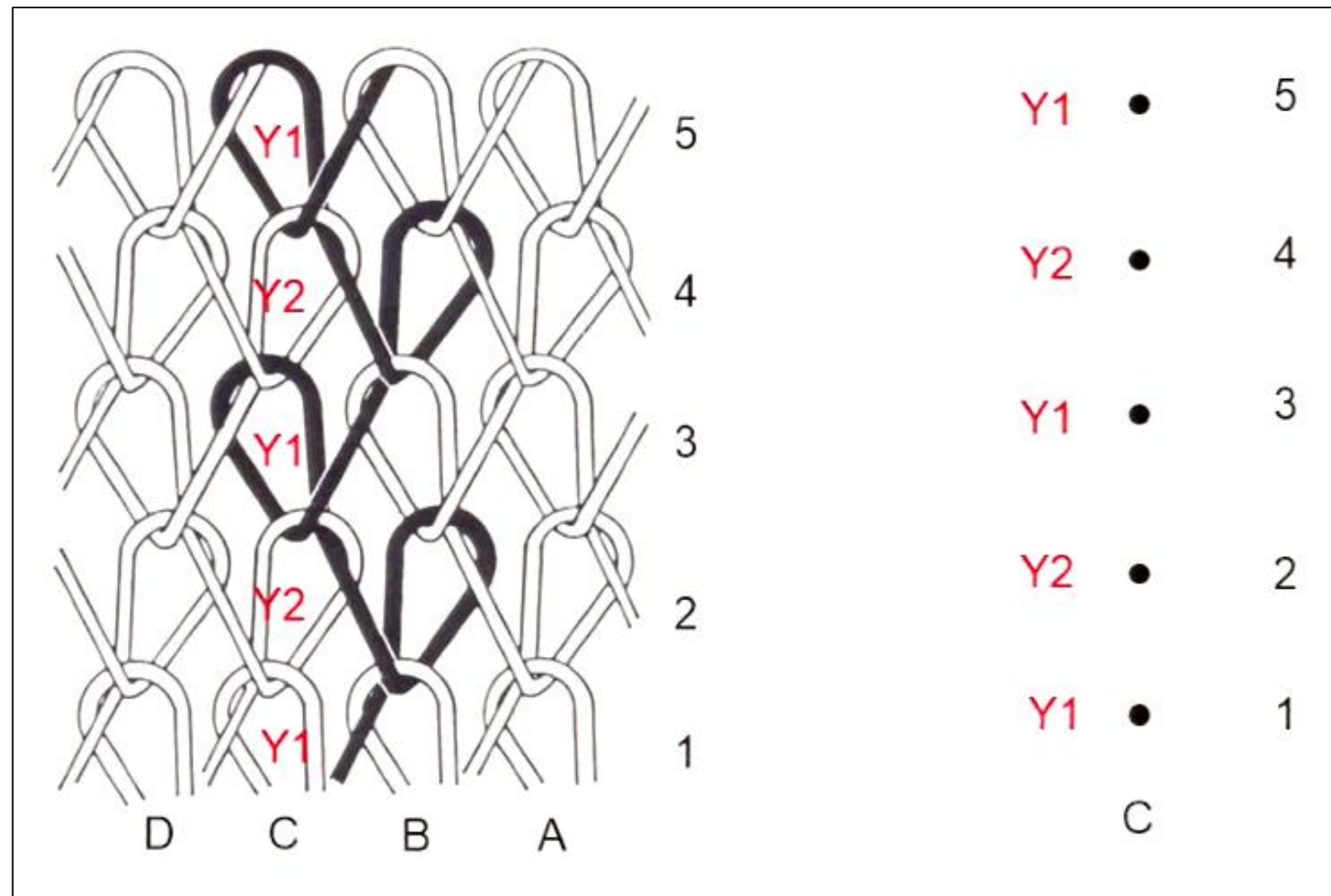


LOOP FORMATION



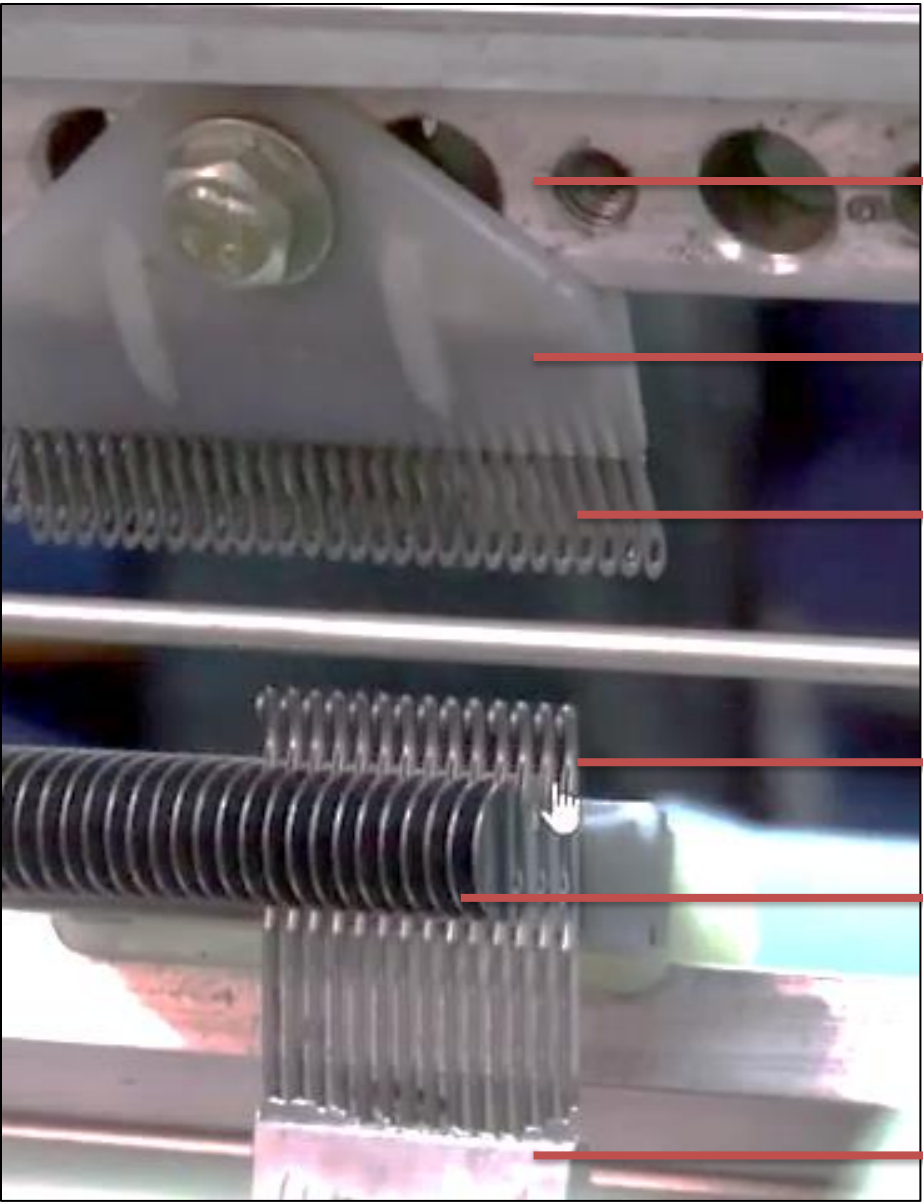
A **Guide (G)** provides **Yarn (Y)** to **Needle (N)** for **Loop** formation

LOOP FORMATION



Guide G2 first supplies yarn to needle N2 in the 1st course and then moves to supply yarn to needle N3 in the 2nd course. This demonstrates how guides change courses and supply yarn to different needles.

WARP KNITTING ELEMENTS



Guide Bar

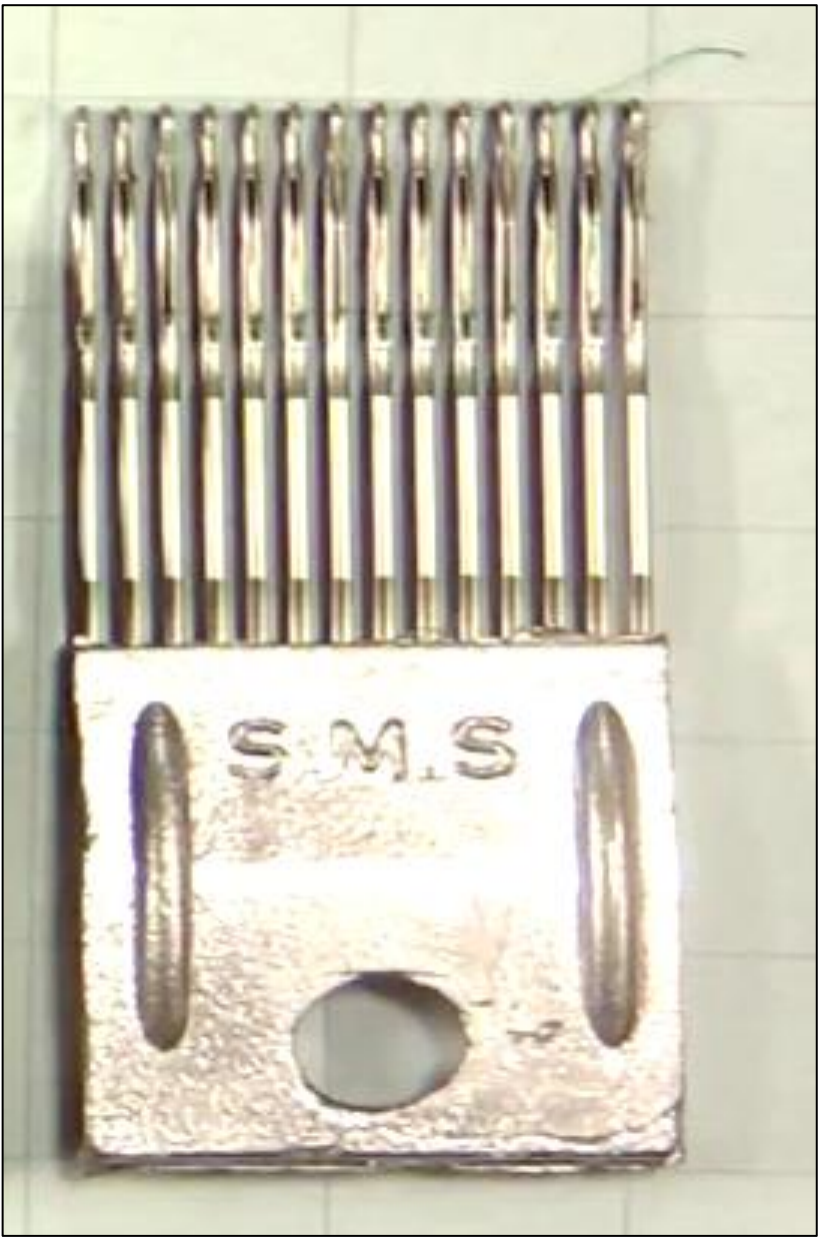
Leads

Guide

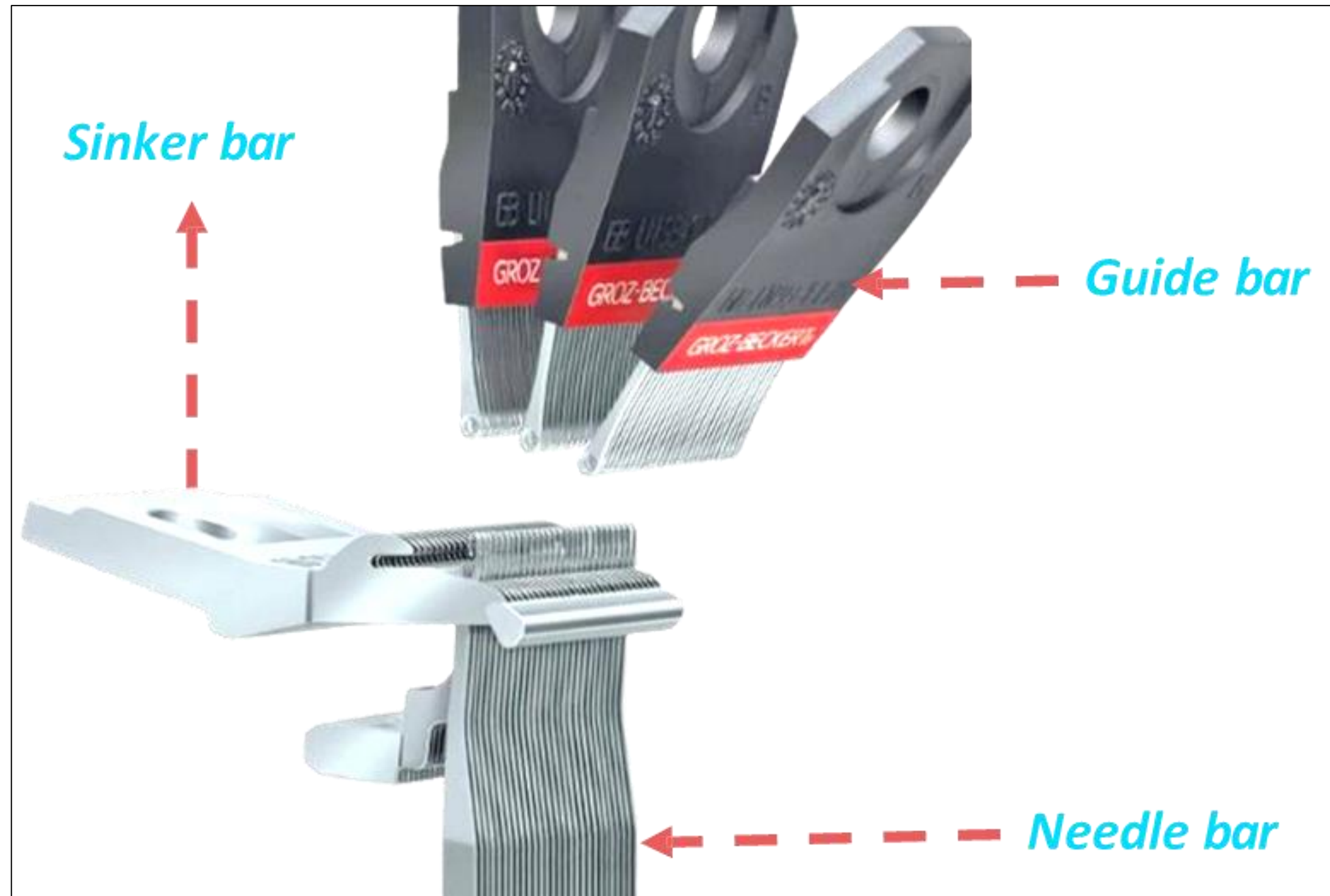
Needle

Sinker

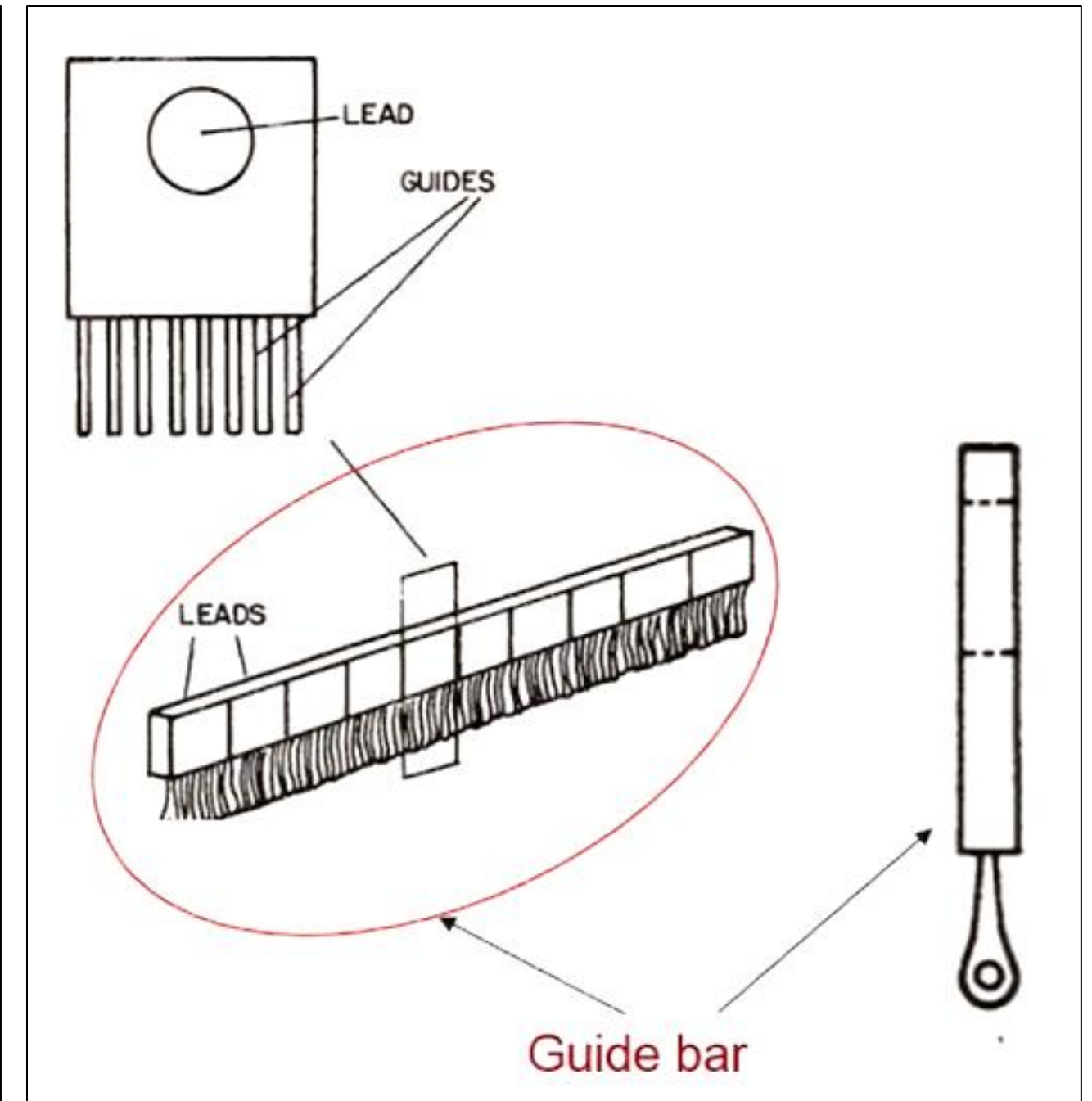
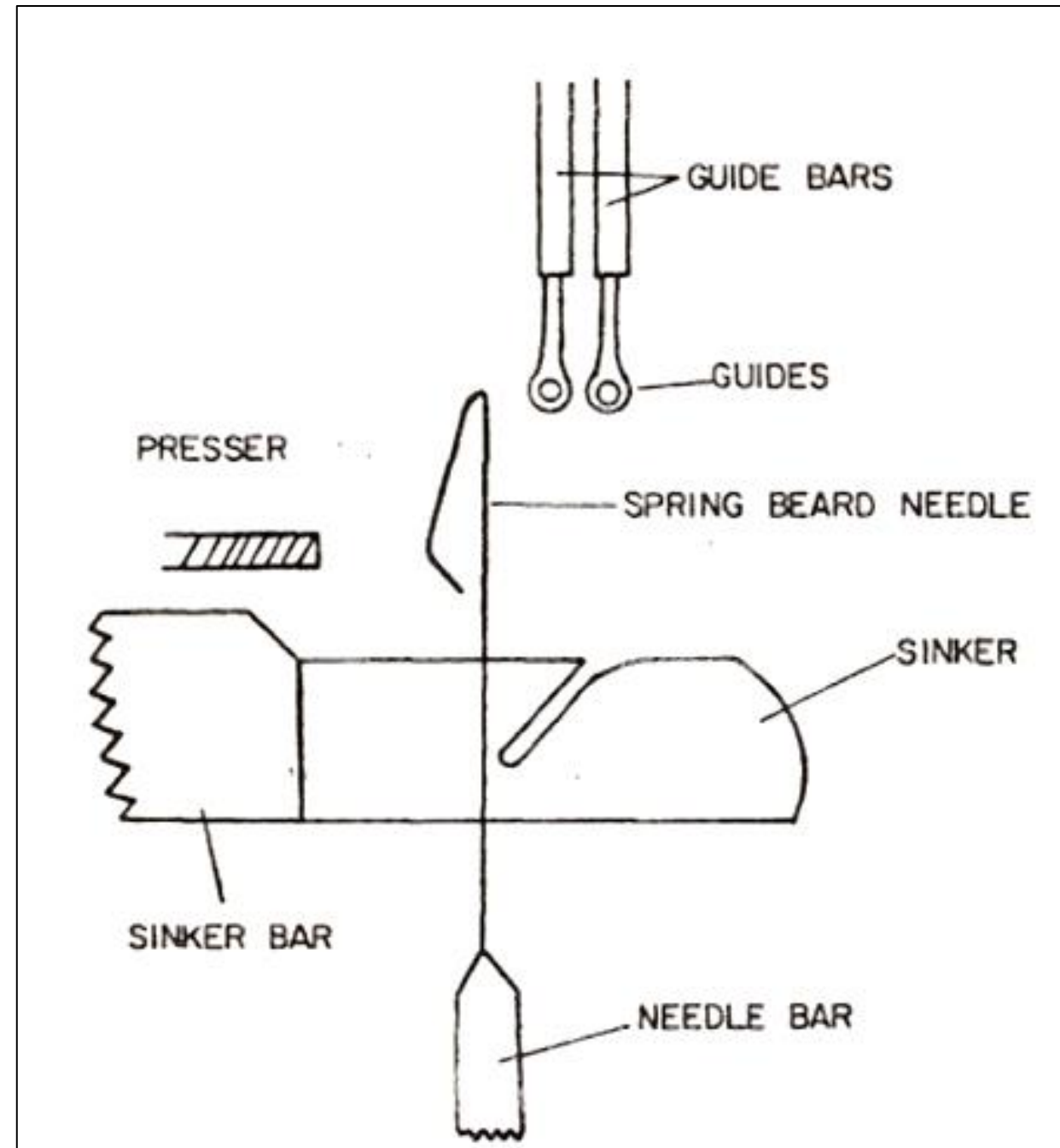
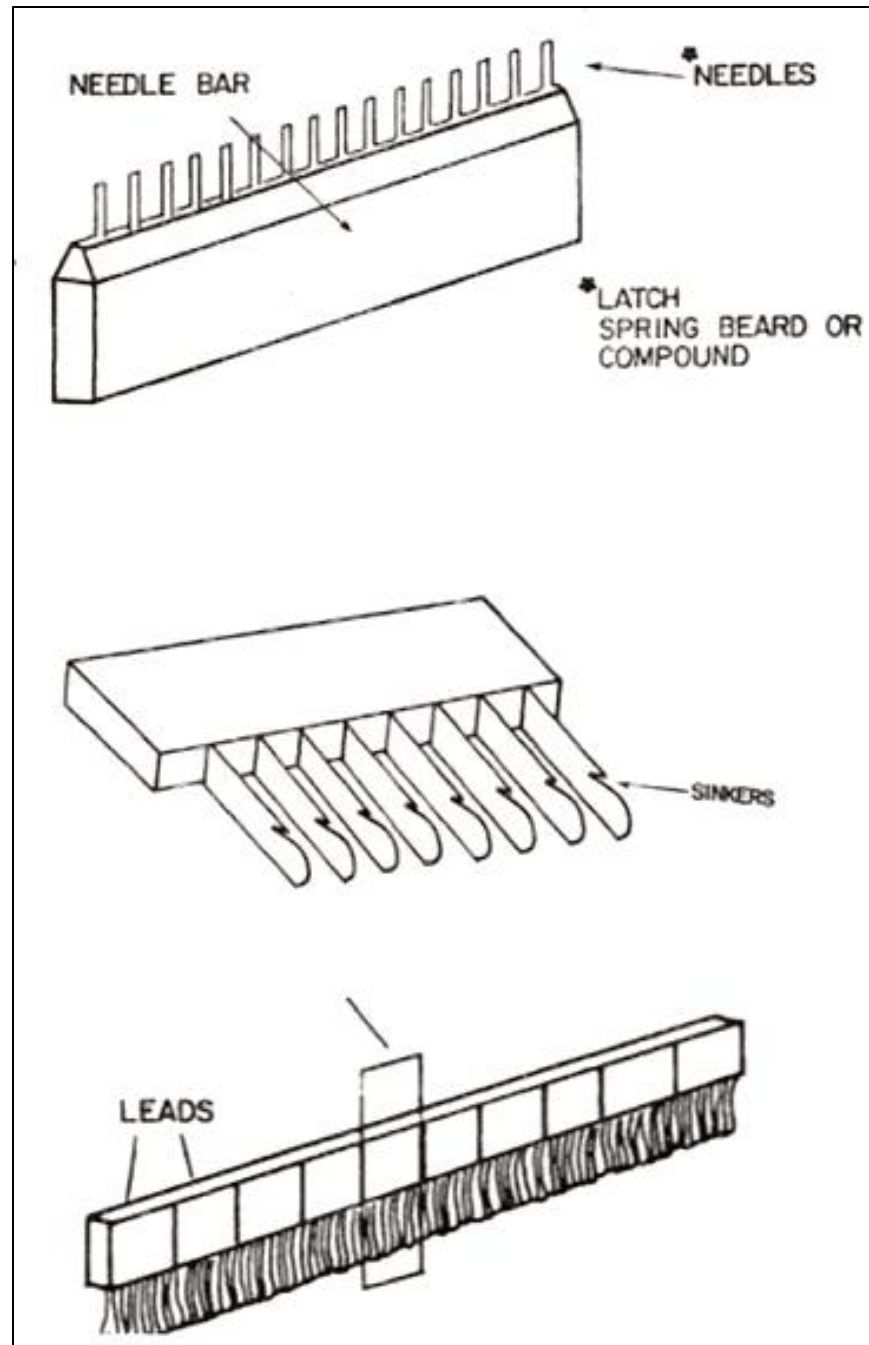
Needle Bar



WARP KNITTING ELEMENTS



WARP KNITTING ELEMENTS



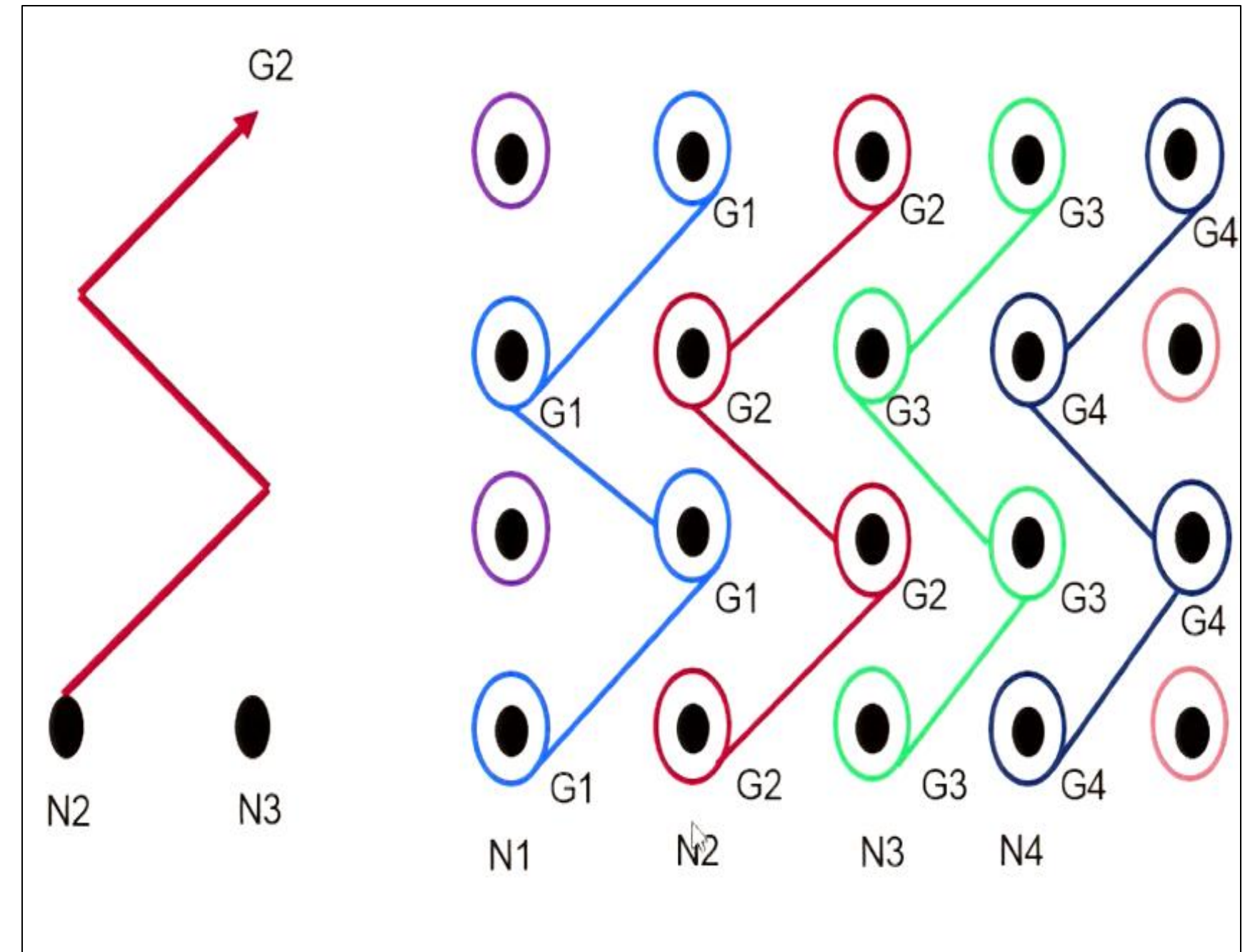
WARP BEAM

To ensure uniform conditions of warp feed and tension, the yarn ends are *supplied from flanged beams* attached to shafts that turn to unwind the warp sheet in parallel formation. For convenience of handling, a number of beams may be attached to a beam shaft *to achieve the full width of the warp sheet*; for example, a warp sheet *84 inches* (213cm) wide might be supplied from a full-width beam, from *two beams each 42 inches* (106cm) wide, or from *four beams each 21 inches* (53cm) wide.



GUIDE

- The guide helps to move the yarn from one needle to another needle.
- Each needle has one guide that supplies yarn.
- The guide controls the yarn and changes its position to supply yarn to different needles in different courses.



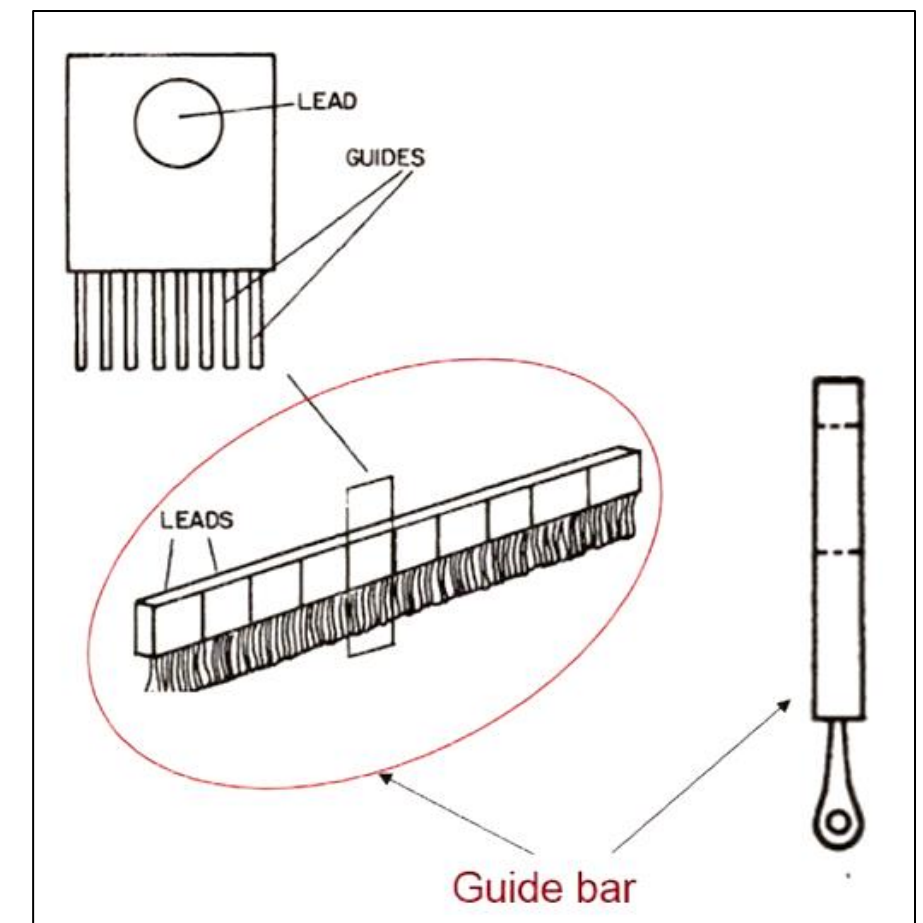
GUIDE

Guide Bar & Guides

- A guide bar is a bar that runs across the full width of the machine.
- It is equipped with guides through which yarns pass. The guides transfer the **side movement from the pattern control device** to the yarns.
- The guides swing between the needles to wrap the yarn and form new loops.
- They also **move sideways (shogging) to connect the wales** and form the fabric.
- Usually, a commercially acceptable warp knitted structure requires a minimum of two guide bars and two warp sheets.

Guides are **thin metal plates with holes** through which yarn passes. They are fixed together in a metal lead and spaced according to the needle gauge.

- Guides hang from the guide bar and move between needles.
- When the guide crosses the needle hook ~ overlap is formed.
- When it moves behind the needle ~ underlap is formed.
- All guides on a guide bar make the same lapping movement, so they require the same yarn tension and feed rate.



SINKER

- The sinker is a thin plate of metal which is placed between every two needles.
- The sinkers are usually cast in units, 1 inch long, which in turn are screwed into a bar called *sinker bar*.
- The sinkers are given almost linear *horizontal* (forward and backward) motion through the sinker bar.
- The *throat of the sinker* are used to hold down the fabric while the *belly of the sinker* is used as a knocking over platform.



LAPPING MOVEMENT OF GUIDE BAR

Single Needle Bar Structure: Lapping Movement of Guide Bar

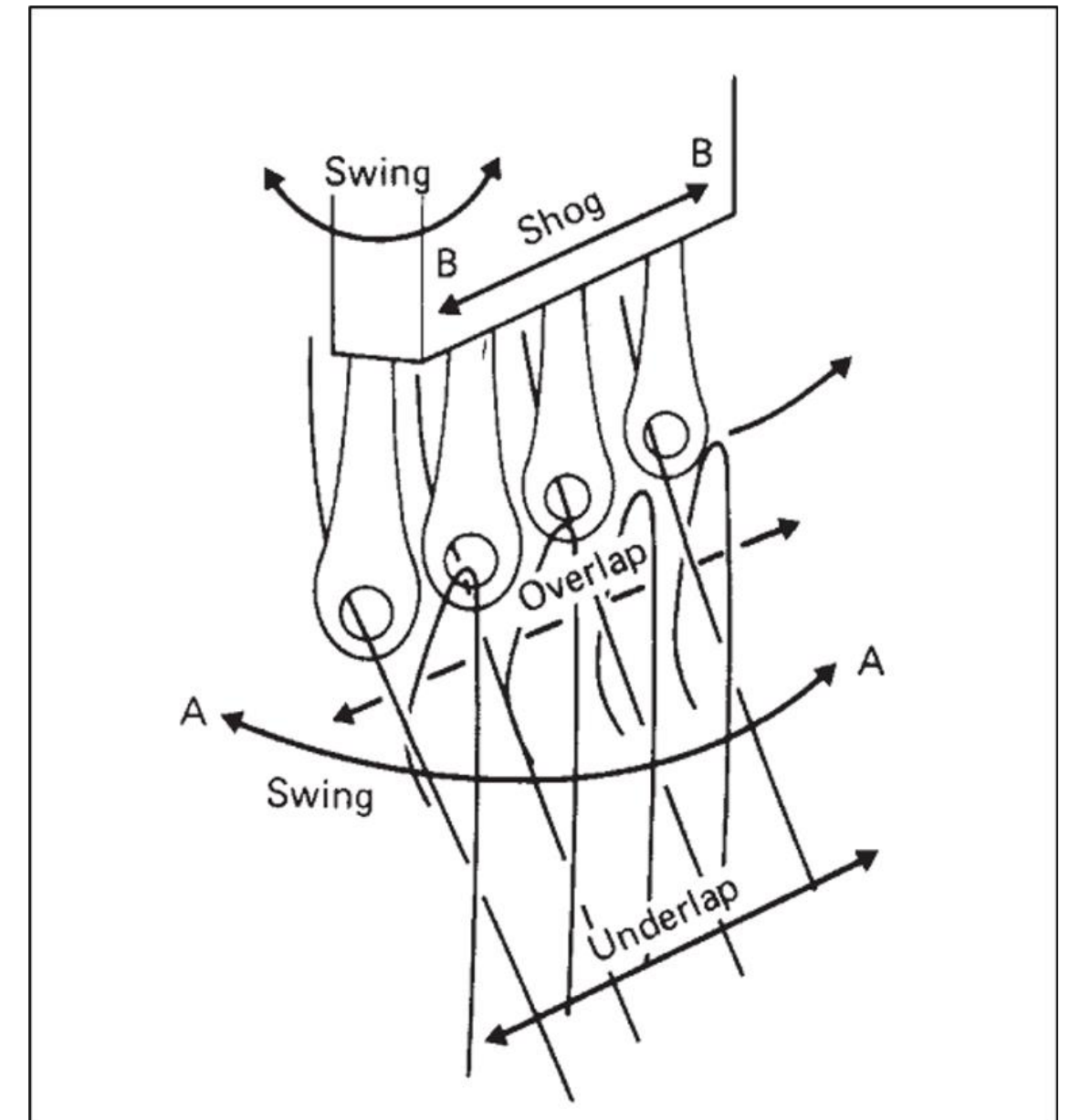
When the needle bar is viewed from the top, the guides of the guide bar perform a compound movement consisting of two separate movements:

Swinging Motion (A–A)

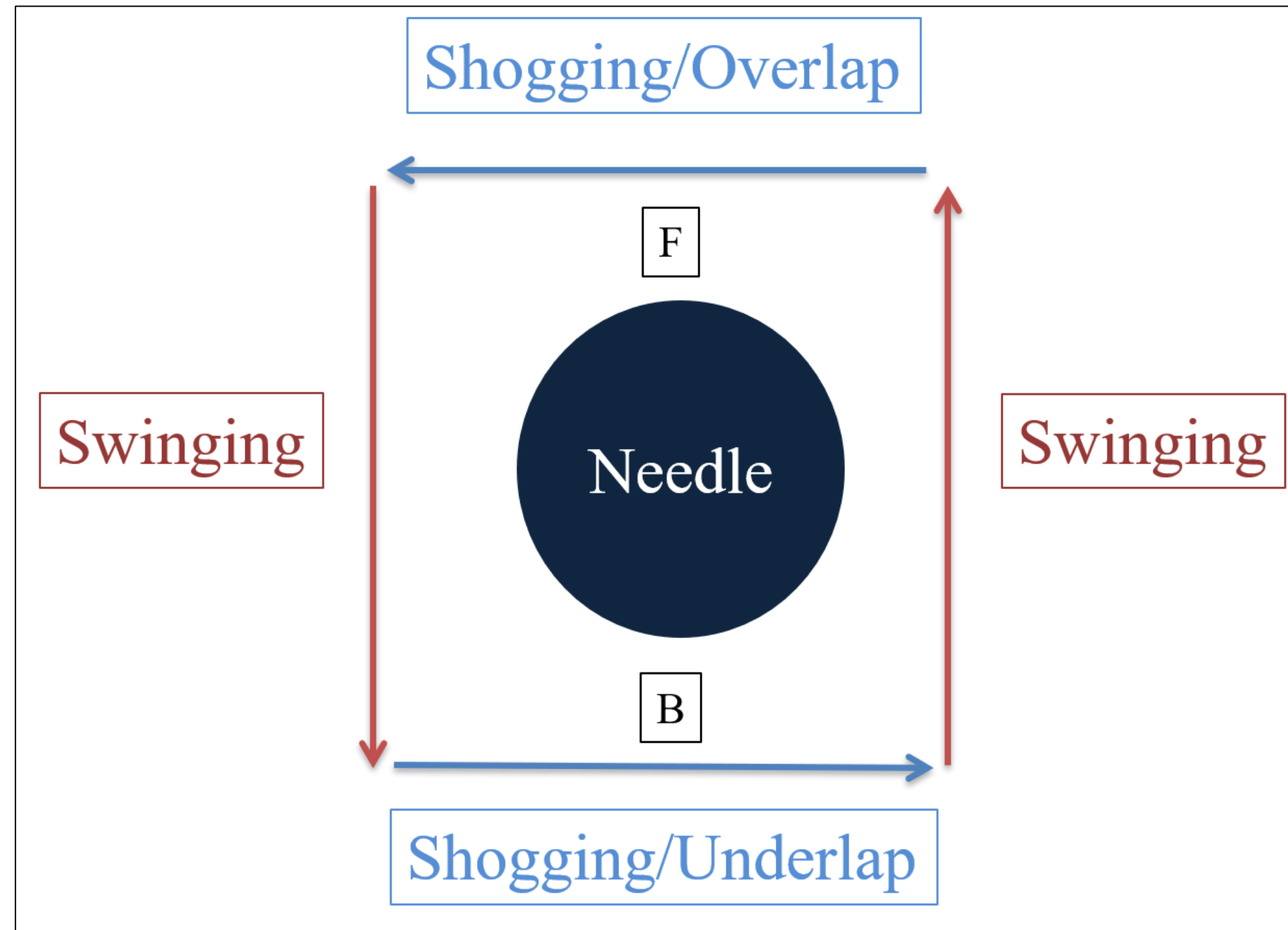
Shogging Movement (B–B)

These two movements **act at right angles to each other**.

Together, they create the *overlap and underlap paths* of the yarn, forming a complete yarn path around the needles.



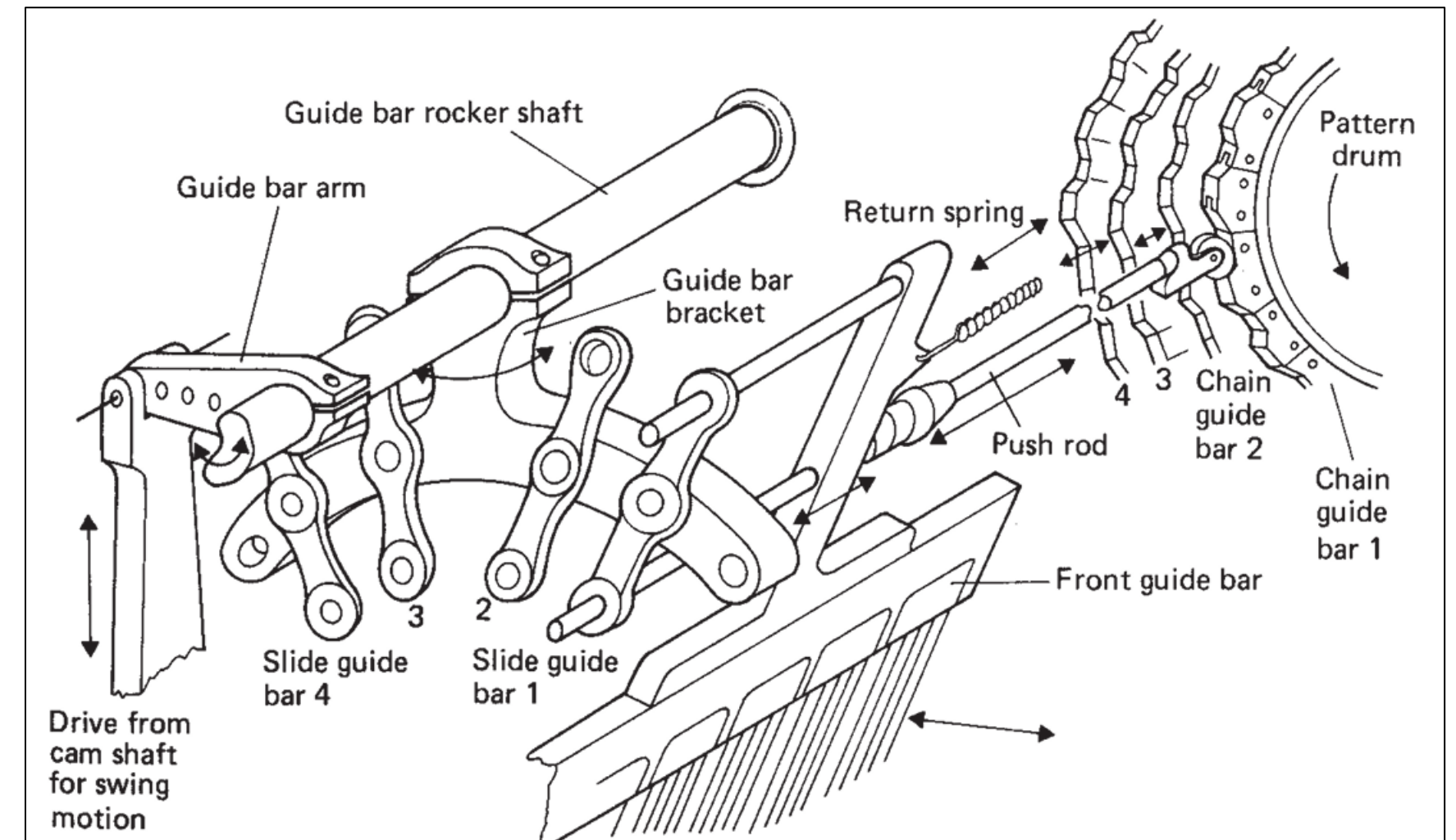
LAPPING MOVEMENT OF GUIDE BAR



LAPPING MOVEMENT OF GUIDE BAR

Swinging Motion

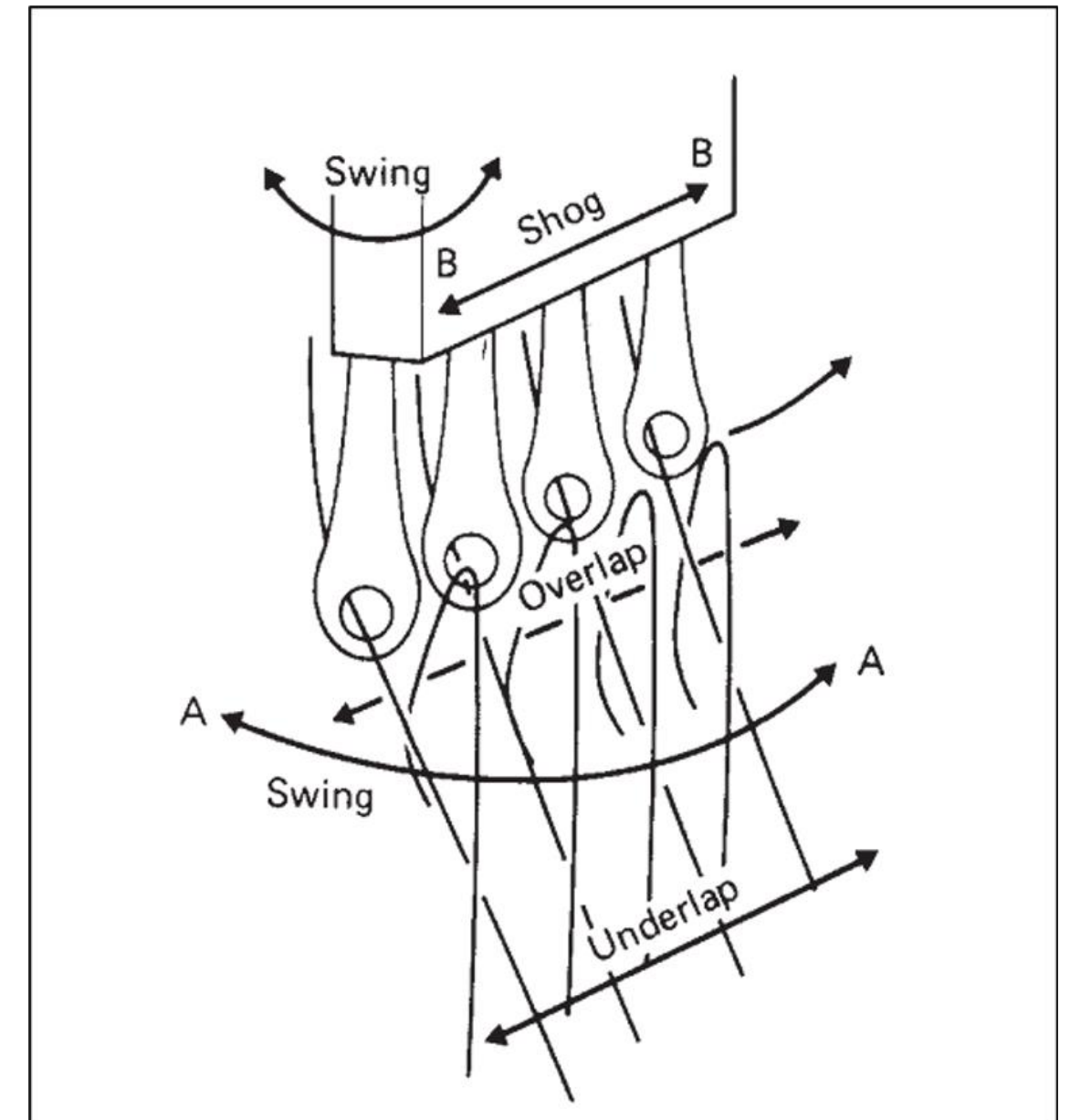
- Swinging motion is the movement of the guide from the *front side of the machine to the hook side of the needle*.
- It happens *between two adjacent needles*.
- This movement is a *fixed* and *automatic* action for all guide bars.
- The guide bars move by rotating around a common rocker shaft.
- Swinging motion *creates the side limbs of the loops*.



LAPPING MOVEMENT OF GUIDE BAR

Shogging Movement

- Shogging is the sideways movement of the guide bar, which occurs parallel to the needle bar.
- This movement creates the overlap and underlap of the yarn.
- The amount, direction, and timing of shogging are controlled separately for each guide bar.
- It is controlled by pattern chain links or pattern wheels.



LAPPING MOVEMENT OF GUIDE BAR

Types of Shogging Movement

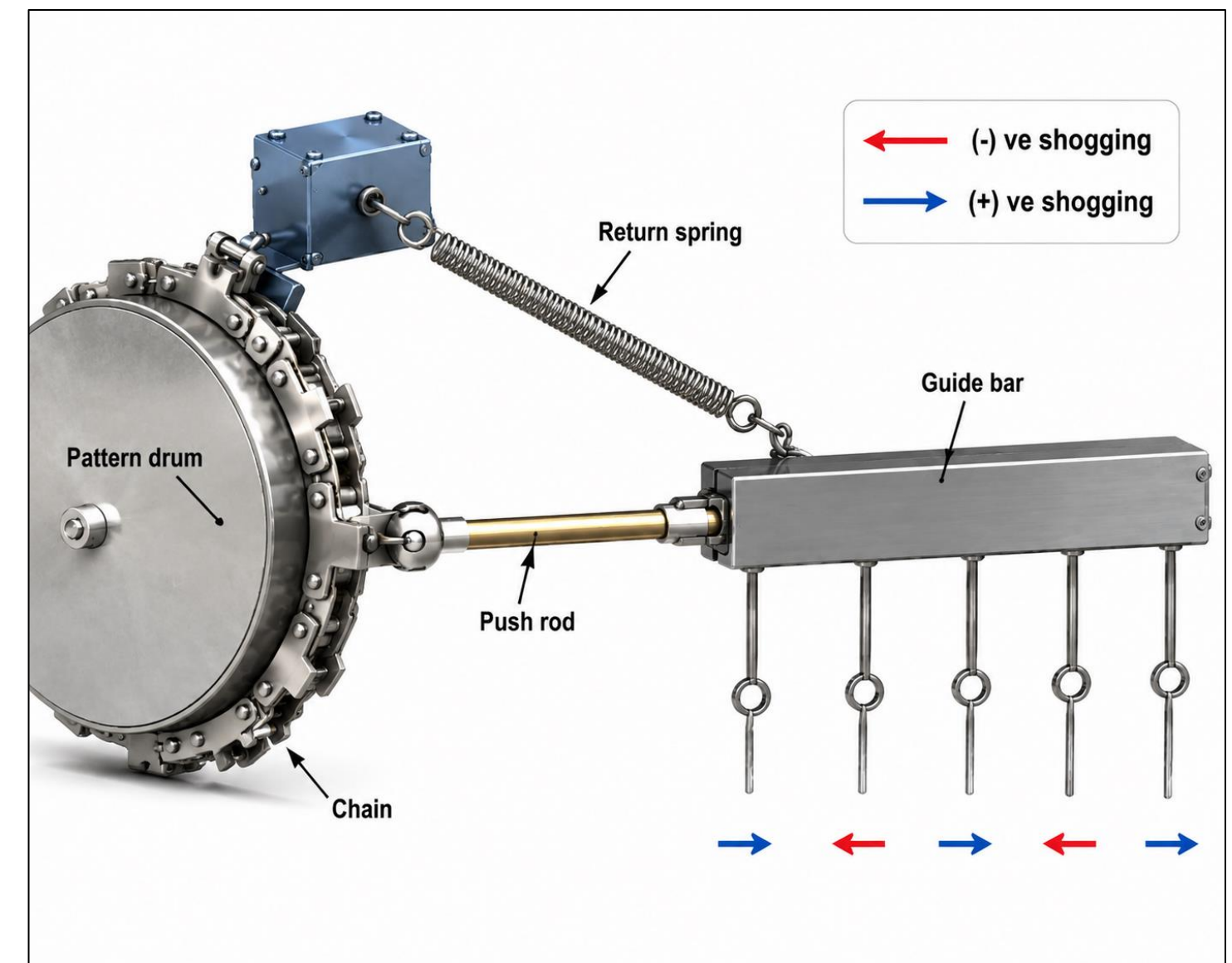
There are two types of shogging movement:

1. Positive Shogging

In positive shogging, the pattern wheel pushes the guide bar away from the pattern mechanism.

2. Negative Shogging

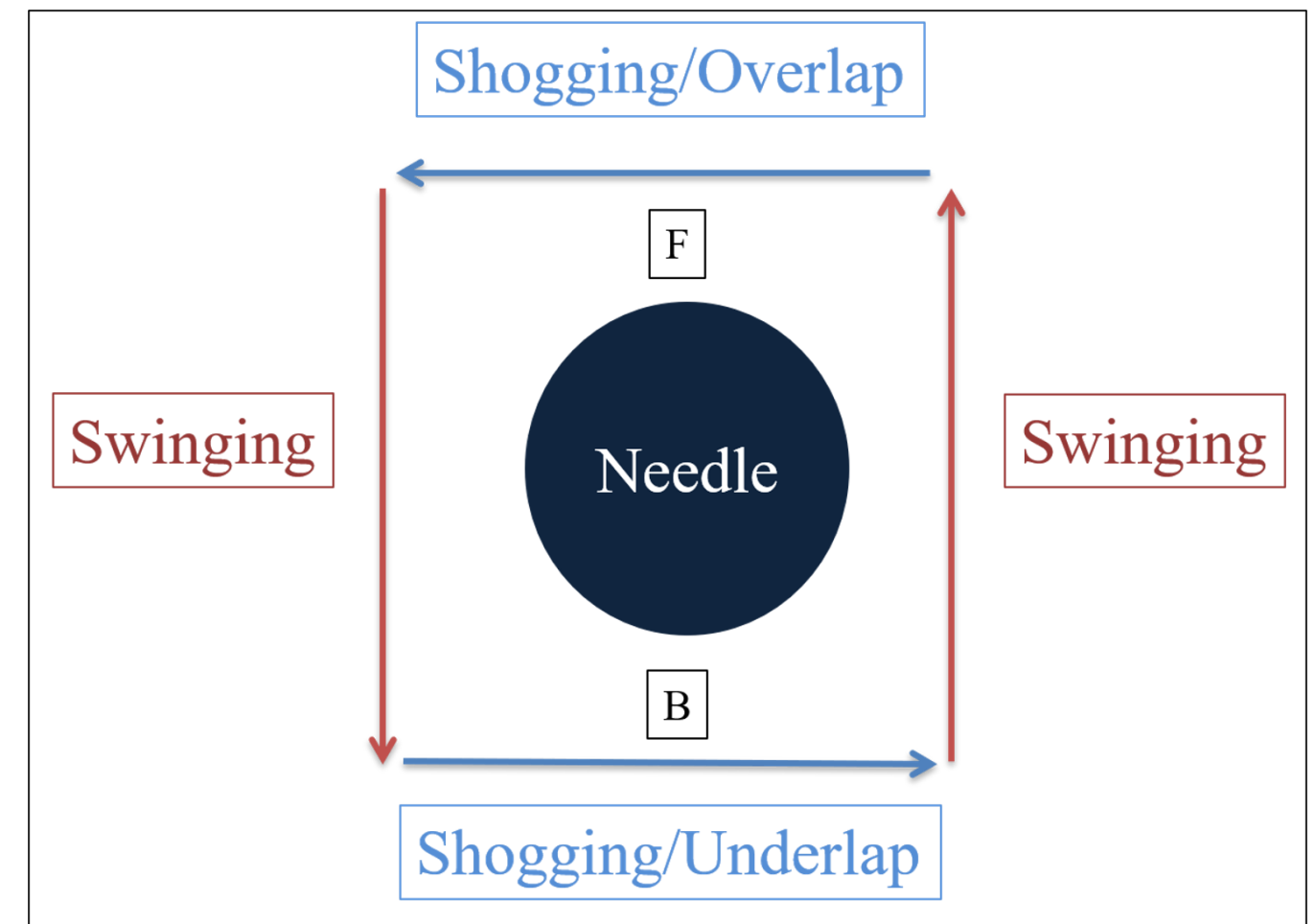
In negative shogging, the pattern wheel pulls the guide bar towards the pattern mechanism.



LAPPING MOVEMENT OF GUIDE BAR

Shogging occurs when the guides move after swinging clear of the needle heads (front or back side of the machine).

- When the guide moves on the hook side (F) of the needle, it produces an overlap.
- When the guide moves away from the hook side, it produces an underlap.



APPLICATION

Industries Apparel

Widely used in lingerie, sportswear, and outerwear.

Home Textiles

Ideal for curtains, upholstery, and bedding.

Automotive Textiles

Used in seat covers, headliners, and trims, these fabrics deliver dimensional stability, wear resistance, and enhanced comfort.

Sports and Outdoor Gear

Warp-knitted fabrics excel in moisture-wicking, stretchability, and durability, making them perfect for performance-driven applications.



THANK YOU !

